Lecture #17: RL-alignment By Percy

obj: $\max_\theta E_{\tau\sim P_\theta(\tau)}[R(\tau)] = \max_\theta \int P_\theta(\tau) R(\tau) d\tau$ policy gradient

目标 = $\sum P_\theta(\tau) \times R(\tau)$ → 好不好 (Reward)

所有轨迹 ← ; ↳ 概率

⇒ Last lecture: overview of RL from verifiable Rewards

⇒ This lecture: deep dive into the mechanics of policy gradient (e.g. GRPO)

RL 就是用 reward 加权轨迹概率, 然后通过梯度, 把概率质量推向高 Reward 轨迹

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① rl_setup_for_lanuage_models()
② policy_gradient()
③ training_walkthrough()

gradient ↓ policy ↳ high reward → (↑) 轨迹 → 概率 高 (↑)

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①: rl_setup_for_lanuage_models()

state ⇒ s: prompt + generated response so fas

action ⇒ a: generate a new token.

Rewards ⇒ R: how good the responses are

sparse reward, high variance

↓ outcome rewards: how good is the whole response / subjective : {PPO, GRPO}

↓ verifiable rewards: objectively ~~ob~~ check the answer / ~~objective~~ / binary / low noise

{ ~~determ~~ deterministic / clearance / supervised

lower variance: faster convergence

{DAPO, ~~G~~SPO, GDSPO}

↓

Notions of discounting and bootstrapping are less appliable

Trasition probabilities: $T(s'|s,a)$: deterministic $s' = s + a$

⇒ can do planning / test-time compute (unlike in robotics)

⇒ states are really made up (different from robotics) with flexibility

Policy: $\pi(a|s)$: just a language model (fine-tuned)

Rollout / episode / trajectory: $s \to a_1 \to a_2 \to a_3 \cdots \to a_n \to R$.

objective: maximize expected reward $E(R)$: expectation = prompt (s) responses (a)